Lecture 10 2025-10-02

Thm 2.6 $f \in C^2([a,b])$ $p \in (a,b)$ with $f(p)=0$, $f'(p) \neq 0$

Then $\exists \delta > 0$ s.t. $p_n \to p$ if $p_0 \in [p-\delta, p+\delta]$.

Proof: Idea: Recast as a fixed point iteration $g(p)=p$

$$g(x) = x - \frac{f(x)}{f'(x)}$$

$$g'(x) = \frac{f(x)f''(x)}{(f'(x))^2}$$

If $f(p)=0 \Rightarrow g'(p)=0$ (0)

Since $f \in C^2([a,b])$, g' is continuous $\overset{(0)}{\Rightarrow}$ g' is small near p

$\Rightarrow$ fixed point iteration converges on some interval containing p.

Goal: Choose $\delta > 0$ small enough s.t.

① $[p-\delta, p+\delta] \subset [a,b]$

② $|f'(x)| \neq 0$ for $|x-p| < \delta$

③ $|g'(x)| < \frac{1}{2}$ for $|x-p| < \delta$

① is possible $p \in (a,b)$ (just set $\delta = \min(p-a, b-p)$)

② Let $\varepsilon = \frac{|f'(p)|}{2} > 0$. Decrease δ if necessary so that

if $|x-p| < \delta \Rightarrow |f'(x) - f'(p)| < \varepsilon$ (possible because f' is continuous)

$$0 < 2\varepsilon = |f'(p)| \overset{\Delta\text{-inequality}}{\leq} |f'(p) - f'(x)| + |f'(x)|$$

$$< \quad \varepsilon \quad + |f'(x)|$$

Subtract ε on both sides $\Rightarrow$

$$0 < \varepsilon < |f'(x)| \quad \text{for} \quad |x-p|<\delta \Rightarrow f'(x) \neq 0$$

③ $g'(x)$ is continuous and $g'(p)=0$, so we can again decrease δ if needed so that

$$|g'(x)-\underbrace{g'(p)}_{=0}| < \frac{1}{2} \quad \text{if } |x-p|<\delta \quad \text{(since } g'(x) \text{ is continuous)}$$

$$\Rightarrow |g'(x)| < \frac{1}{2} \quad \text{if } |x-p|<\delta.$$

<u>Claim</u>: g maps $[p-\delta, p+\delta]$ to itself

$$|g(x)-\underbrace{g(p)}_{=p}| = \underbrace{|g'(\xi)}_{|g'(\xi)|<\frac{1}{2}\ \xi\in[p-\delta,p+\delta]}|(x-p)| \leq \frac{1}{2}|x-p| \leq \frac{\delta}{2} < \delta$$

$$\Rightarrow |g(x)-p| < \delta \Rightarrow g(x) \in [p-\delta, p+\delta]$$

Therefore the hypotheses of the fixed point theorem are satisfied and $p_n = \underbrace{g(p_{n-1})}_{p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})}}$ converges to p

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Secant Method

Newton's Method is very powerfull but has a weakness: We need to know the derivative of f!

<u>Idea</u>: Replace the derivative by a finite difference

approximation:

We know $$f'(p_n) = \lim_{x \to p_n} \frac{f(x) - f(p_n)}{x - p_n}$$

If p_n and p_{n-1} are close then

$$f'(p_n) \approx m = \frac{f(p_n) - f(p_{n-1})}{p_n - p_{n-1}}$$

So if we replace $f'(p_n)$ by m in Newton's update, we obtain

$$p_{n+1} = p_n - \frac{(p_n - p_{n-1})\, f(p_n)}{f(p_n) - f(p_{n-1})}$$

↝ Need two initial guesses p_0, p_1.

Algorithm 2.4 (Secant method)

Input: f, initial guesses p_0, p_1, tolerance ε, maximum number of iterations N_{max}

Output: Approximation of p (root of f) or message of failure

Step 1: Set $i = 2$

$q_0 = f(p_0)$

$q_1 = f(p_1)$

Step 2: while $i \leq N_{max}$ do

* Set $p = p_1 - \frac{q_1(p_1 - p_0)}{q_1 - q_0}$ (compute p_i)

* If $|p - p_1| < \varepsilon$ then

return p;

* Set $i = i + 1$

* Set $p_0 = p_1$

$q_0 = q_1$

$p_1 = p$

$q_1 = f(p)$

Step 3: return "The method failed after N_{max} iteration"

* Newton's Method sometimes fails to converge:

$f(x) = \tan^{-1} x \qquad f'(x) = \frac{1}{1+x^2}$

$p_{n+1} = -g(p_n) \qquad g(x) = \underbrace{(1+x^2)\tan^{-1} x}_{\frac{f(x)}{f'(x)}} - x$

$q_n = (-1)^n p_n$

$q_{n+1} = (-1)^{n+1} p_{n+1} = (-1)^n g(p_n) = g((-1)^n p_n) = g(q_n)$

(g is an odd function, n is odd)

g has 2 fixed points $x = 0$, $x = p = 1.391745$

If p_0 starts at p, it will cycle between $\pm p$

$p_1 = -g(p_0) = -p_0$

$p_2 = -g(p_1) = g(-p_1) = g(p_0) = p_0$ (g odd)

$p_n = (-1)^n p_0$ (while q_n stays fixed at p_0)

If $|p_0| < p$: $p_n \to 0$ and alternates in sign

$q_n \to 0$ monotonically ($q_n = (-1)^n p_n$ if $p_0 < 0$)

If $|p_0| > p$: p_n grows in magnitude and alternates sign

q_n grows monotonically

2.4. Order of Convergence (for iterative methods)

Def. 2.7 Suppose $p_n \to p$ (and $p_n \neq p$ for $n \geq N$)

We say that $p_n \to p$ with order α if $\exists \lambda$ such that

$$\lim_{n \to \infty} \frac{|p_{n+1} - p|}{|p_n - p|^\alpha} = \lambda = \text{asymptotic error constant.}$$

Linear convergence: $\alpha = 1$, $\lambda < 1$

Quadratic convergence: $\alpha = 2$